

**ESTÁCIO UNIVERSITY CENTER
GOIANA – PE DISTANCE LEARNING HUB**

TECHNICAL REPORT FOR THE DATA SCIENCE COURSE

**Predictive Model for Magazine Luiza's (MGLU3) Revenue Growth: A
Critical Analysis of Linear Extrapolation in E-commerce.**

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ABSTRACT

E-commerce, as a growing force both globally and nationally, has an extreme need for data collection and analysis. In this analysis, we continue experimenting with extrapolation models for future business decision-making, focusing on Magazine Luiza.

The models applied were linear regression and non-linear regression, more specifically Random Forest.

Throughout the analysis, we demonstrate the use of the CRISP-DM methodology, from the selection of the dataset to the implementation of the most valid model for the most realistic prediction of the data, ensuring reliability and robustness.

Keywords: CRISP-DM; Predictive Models; Magazine Luiza; MGLU3; Machine Learning; Time Series.

1. CONTEXT AND OBJECTIVE

This technical report documents the initial stages of the Magazine Luiza (MGLU3) Annual Net Revenue Growth Forecast project for the 2020 to 2030 horizon. The work follows the CRISP-DM methodology and uses historical data from MGLU3 from 2011 to 2019, combined with Brazilian macroeconomic indicators from 2010 to 2020, to build a predictive model.

The main objective is to assess the sensitivity of MGLU3's revenue growth by examining internal variables (Gross Margin, Return on Equity, Asset Turnover) and external variables (GDP, Household Consumption), and finally, to project future growth.

2. METHODOLOGY

CRISP-DM, as the most widely used methodology for Data Mining or Data Science projects, was chosen for this project.

2.1. Methodology Application

Data from Magazine Luiza's e-commerce balance sheet was selected, and to provide greater context and assertiveness, national economic indicator data was also gathered.

The Python language was used as a tool for manipulation and script creation.

2.1.1. Data Preparation

Files were standardized by converting them to CSV format, key financial indicators were calculated from a quarterly to an annual basis, and temporal standardization and granularity were performed with macroeconomic indicators to ensure correct correlation in predictive modeling, concluding with data cleaning to remove empty values.

BEFORE

	A	B	C	D
1	XLSWrite 1.0.4 Copyright(c) 1999.2000 Axolot Data	Balanço Patrimonial - MAGAZ LUIZA		
2		30/06/2020	31/03/2020	31/12/2019
3	Ativo Total	20.558.633	17.710.082	19.791.071
4	Ativo Circulante	13.278.766	10.589.868	12.841.161
5	Caixa e Equivalentes de Caixa	1.103.523	388.904	305.746

BEFORE

Zero values

9	Ativos Biológicos	0	0	0	0
10	Tributos a Recuperar	18.944	877.448	864.144	745.693
11	Despesas Antecipadas	0	0	0	0
12	Outros Ativos Circulantes	963.199	869.713	506.316	384.307
13	Ativo Realizável a Longo Prazo	1.917.940	1.861.255	1.748.703	1.856.321
14	Aplicações Financeiras Avaliadas a Valor Justo	0	0	214	264
15	Aplicações Financeiras Avaliadas ao Custo Amortizado	0	0	0	0
16	Contas a Receber	0	0	0	0

Tabela 2. Taxas de crescimento do PIB (2011/2020).

Anos	Variação real	PIB per capita
2011	3,97	3,07
2012	1,92	1,04
2013	3,00	2,13
2014	0,50	-0,35
2015	-3,55	-4,38
2016	-3,28	-4,07
2017	1,32	0,52
2018	1,78	0,96
2019	1,22	0,40
2020	-3,88	-4,60
Média anual	0,26	-0,56
Crescimento acumulado	2,67	-5,50

PDF
spreadsheet

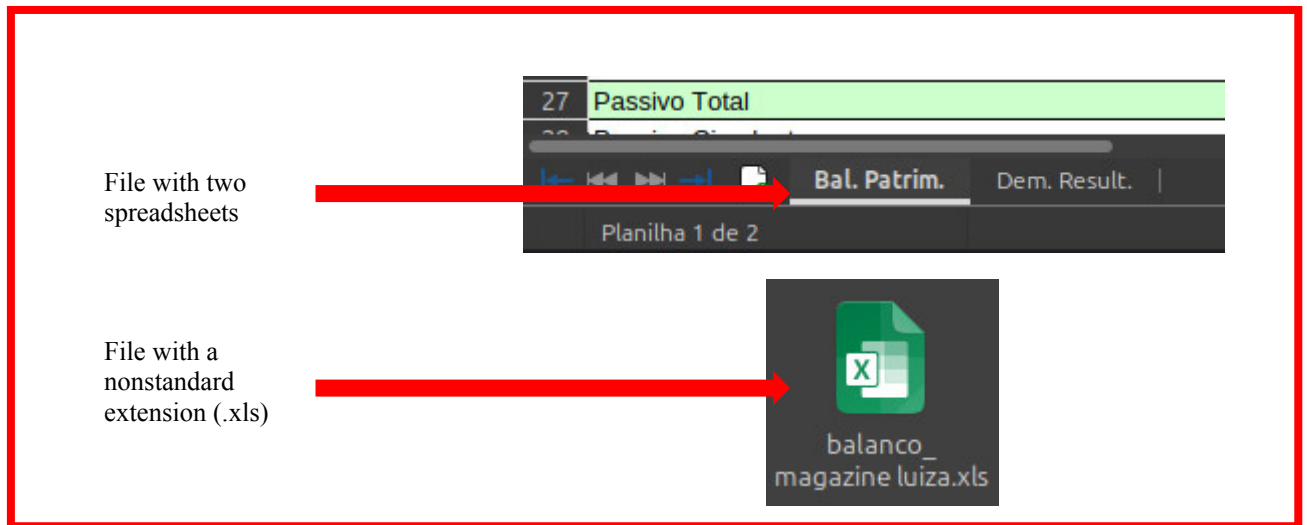
Fonte: IBGE (2022c; 2022d).

AFTER

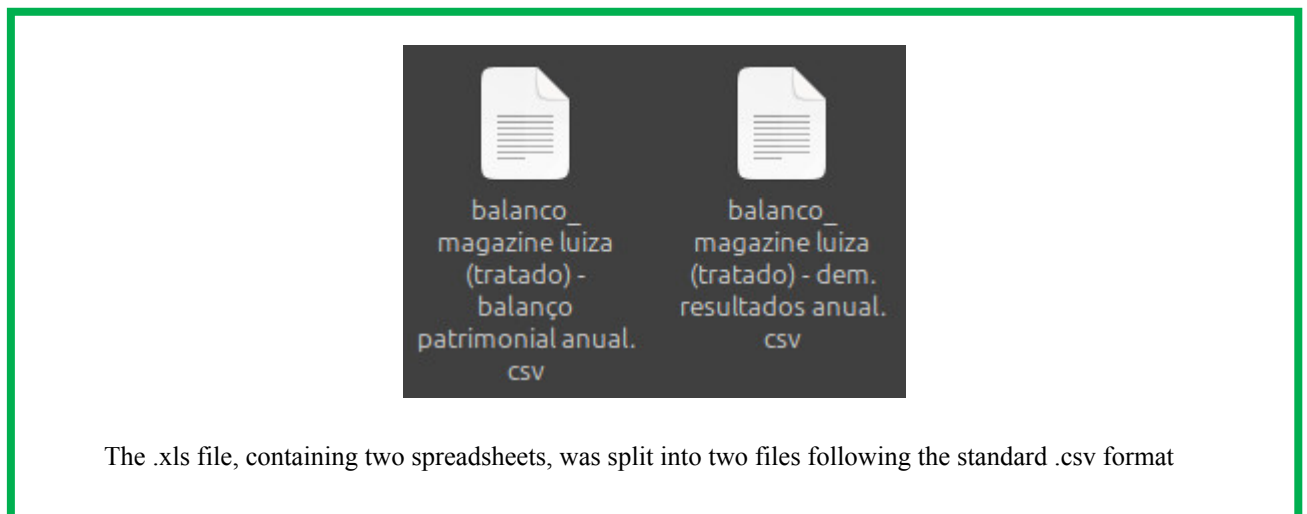
	A	B	C	D	E	F	G	H	I	J
1	Ano Vigente	2019	2018	2017	2016	2015	2014	2013	2012	2011
2	Receita Líquida de Vendas e/ou Serviços	19886309	15590444	11984250	9508746	8978259	9779385	22172922	7665112	6419371
3	Custo de Bens e/ou Serviços Vendidos	-14332350	-11053022	-8378239	-6586130	-6399630	-7086909	-15247125	-5146597	-4275084
4	Resultado Bruto	5553961	4537422	3606011	2922615	2578629	2692476	6925797	2518515	2144287
5	Despesas Com Vendas	-3444112	-2747447	-2119953	-1776258	-1720799	-1746258	-4365789	-1581049	-1270971
6	Despesas Gerais e Administrativas	-1188562	-759833	-679085	-615545	-584280	-556882	-1429211	-481965	-441566
7	Perdas pela Não Recuperabilidade de Ativos	-75993	-59737	-41921	-26074	-30462	-22547	-604662	-338519	-244931
8	Outras Receitas Operacionais	416662	53389	36486	13505	20233	24519	156292	31283	26841
9	Resultado da Equivalência Patrimonial	26607	57757	86156	62702	75605	99620	54464	0	0
10	Financeiras	-70413	-294688	-410821	-503849	-486055	-360742	-581982	-172297	-165723
11	Receitas Financeiras	640917	133929	110107	116655	130297	96469	176779	55695	51686
12	Despesas Financeiras	-711330	-428617	-520928	-620504	-616352	-457211	-758761	-227992	-217409
13	Resultado Antes Tributação/Participações	1218150	786863	476873	77096	-147129	130186	154909	-24032	47937
14	Lucro/Prejuízo do Período	921828	597429	389022	86565	-65605	128556	118727	-6745	11666

A cohesively structured spreadsheet with annual date format, rows containing zero values and irrelevant information excluded, and standardized data in .csv format.

BEFORE



AFTER

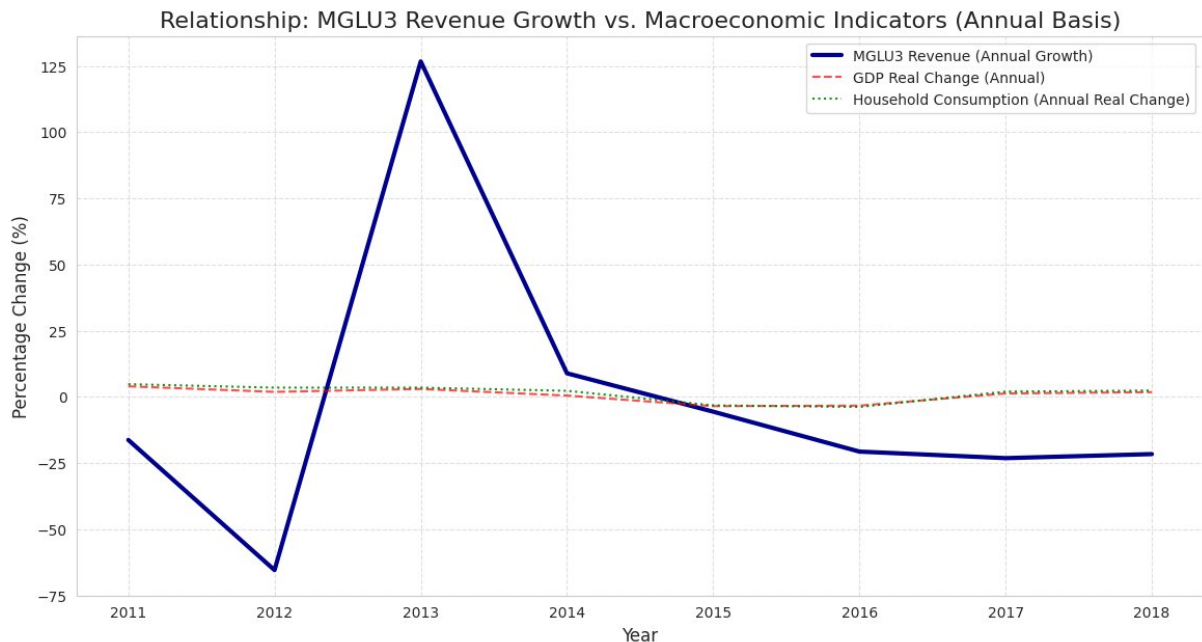


3. PREDICTIVE MODELING

The modeling stage began with the choice to apply a Multiple Linear Regression model to establish predictive correlations. The main reasons for this choice were that this model is a reference in forecasting projects, has a closed and extremely efficient mathematical formulation. It is perfect for the CRISP-DM modeling phase, as it allows the analysis team to quickly perform initial tests, validate correlation hypotheses, and assess the quality of the prepared dataset. Also keeping in mind that basic economic and financial theory suggests that a company's revenue growth has a linear relationship with macroeconomic indicators and internal indicators.

3.1. Key Relationship for E-commerce: MGLU3 vs. Economy

The first script generates a plot to be used in the correlation between MGLU3's (Magazine Luiza) annual revenue growth versus household consumption in the same period.

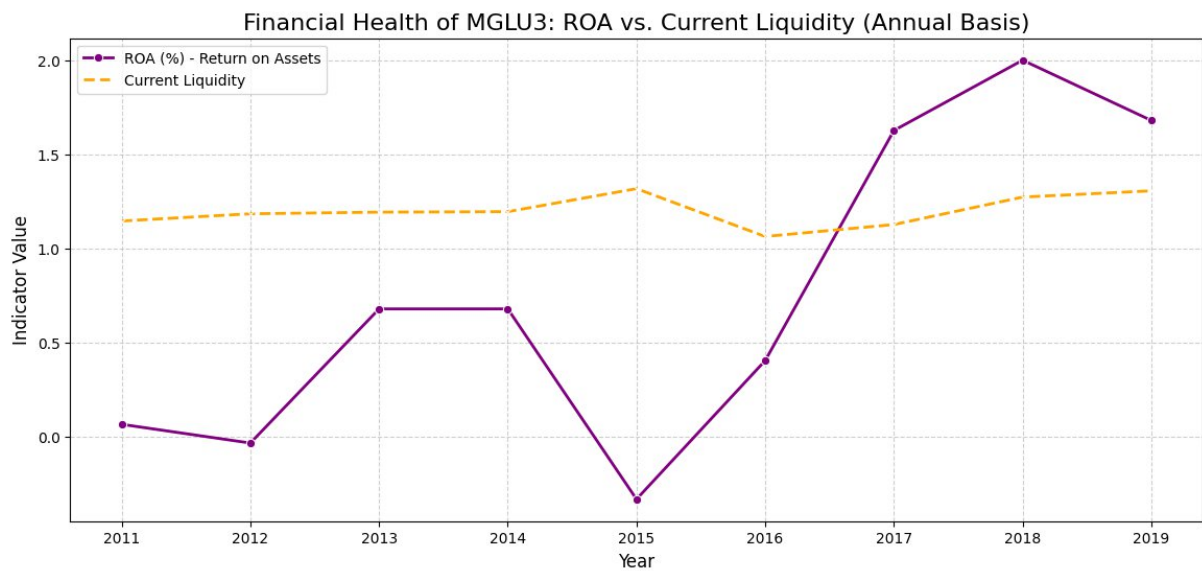


GRAPH 1

The graph illustrates the divergence of Magazine Luiza's revenue growth relative to traditional macroeconomic indicators during the 2011-2014 period, suggesting that growth momentum was driven by internal or sector-specific factors.

3.2. Internal Relationship: MGLU3's Financial Health

The second script generates a plot to show MGLU3's (Magazine Luiza) financial health, observing the company's Return on Assets (ROA) and its current liquidity.



GRAPH 2

The graph shows MGLU3 emerging from a profitability crisis in 2015 to become significantly more efficient and profitable by 2018/2019, while maintaining its financial health (Current Liquidity) at a safe level.

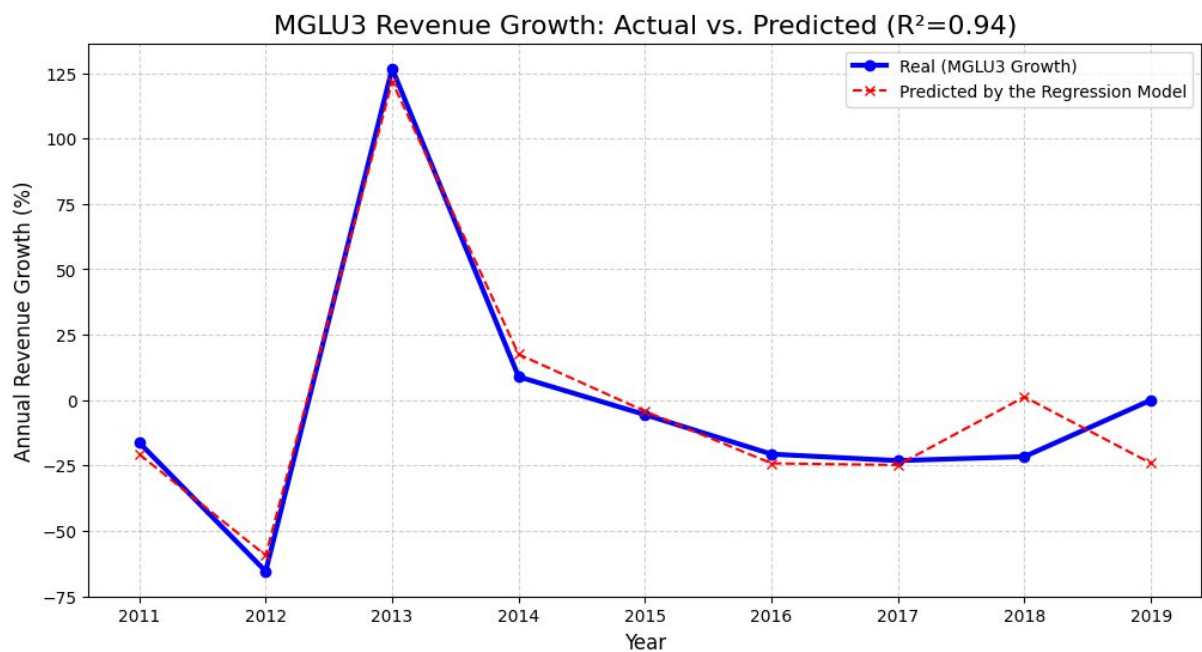
3.3. Building the Final Dataset for Modeling

The third script is responsible for the Extract, Transform, and Load (ETL) process, culminating in the export of a unified and pre-processed dataset for predictive modeling (Machine Learning/AI).

The process integrates three annual data sources – Financial Statements (MGLU3's Income Statement and Balance Sheet) and Macroeconomic Indicators (GDP and Household Consumption) – into a single DataFrame.

3.4. Predictive Modeling (Multiple Linear Regression)

The fourth script generates the plot for the visual representation of the Linear Regression model's performance, observing revenue growth and creating a comparison of the actual and predictive model.

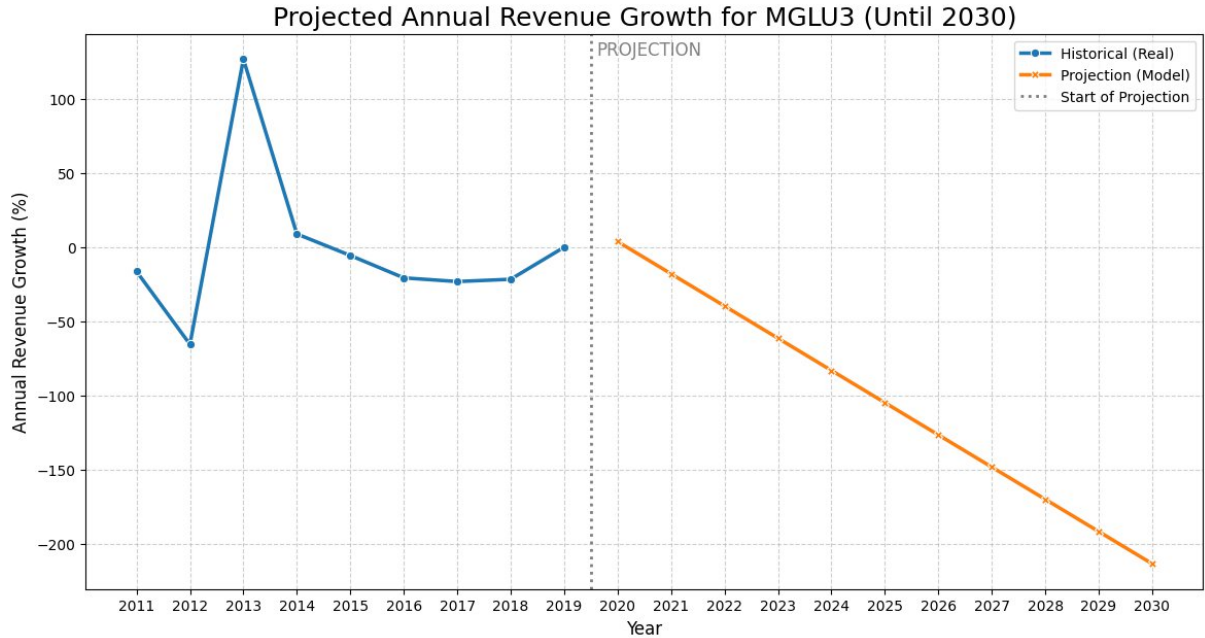


GRAPH 3

The graph demonstrates that the Linear Regression model, using internal and macroeconomic indicators, is extremely robust and explanatory for the target variable "MGLU3 Annual Revenue Growth", with an explanatory power of 94%.

3.4.1. Projection until 2030

The fifth script generates the plot of the future revenue growth trend until 2030.



GRAPH 4

The "MGLU3 Annual Revenue Growth Projection (Until 2030)" graph illustrates the forecast of the developed Linear Regression model, contrasting the real history with the future scenario. It also shows an annual growth trend dramatically falling, reaching -200% in 2030, which is an unrealistic result for any business scenario.

4. DISCUSSION, DIAGNOSIS, AND MODEL LIMITATIONS

Considering the initial objective of evaluating the sensitivity of MGLU3's revenue growth and projecting future growth, we see that the objective was achieved, with some caveats.

4.1. Final Result of the Linear Regression Model

Graph 4 ("MGLU3 Annual Revenue Growth Projection (Until 2030)") shows an unrealistic result given the company's history.

Despite having excellent performance in the historical period (training) with a coefficient of determination (R^2) and an explanatory power of 94%, there is a clear flaw in the future projection, with the probable cause being the linear regression model. Being linear, with a historical series (2011–2019) containing extreme points (peaks and valleys) and a general upward trend in recent years, the model may have adjusted the slope based on initial data, leading to unsustainable extrapolation. The model fails to capture the non-linearity and growth limits of a company.

5. NON-LINEAR MODEL VALIDATION AND FINAL RECOMMENDATIONS

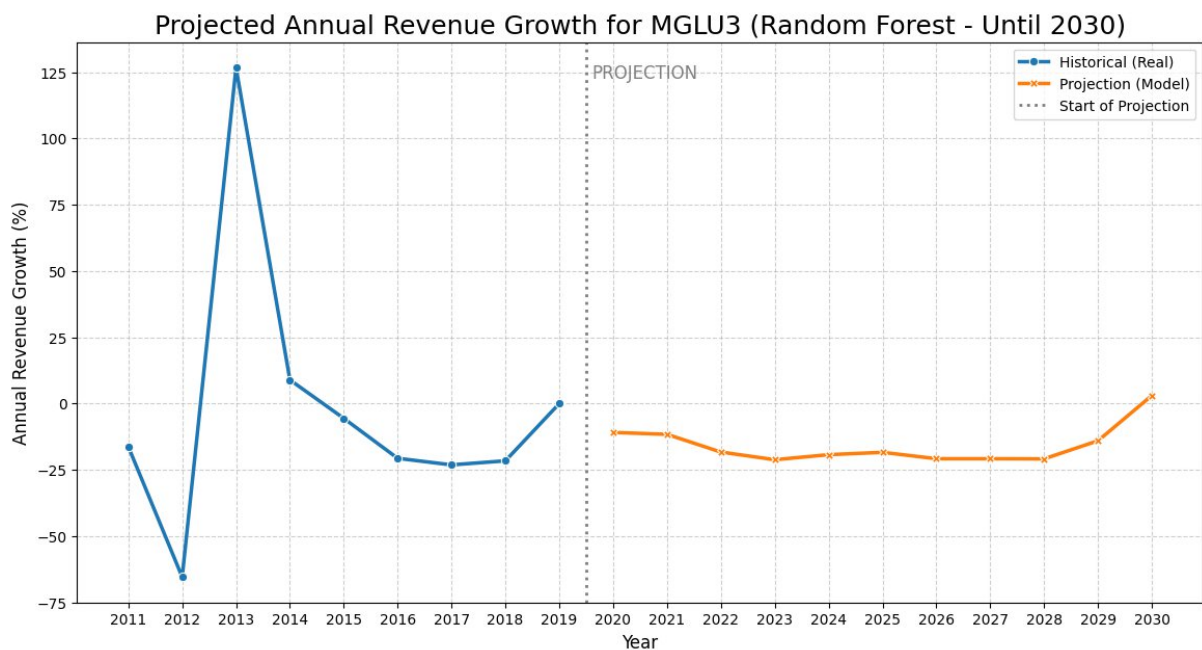
The growth of an e-commerce company is a non-linear phenomenon. Growth is exponential in certain phases, but then stabilizes (Law of Large Numbers). Following this knowledge and given the lack of confidence in the generated future projection, it is advisable to seek alternative models.

Solutions involve shifting focus to more robust models, such as time series, non-linear data, and growth curve models. Some examples of models to pursue with the project are ARIMA/SARIMA, Random Forest Regressor, XGBoost, and Prophet.

The strongest recommendation for the new model is the Random Forest Regressor, to overcome the limitation of extrapolation from the initial linear model.

5.1. Validation with the Random Forest Model

Following the recommendation for a new model, the Random Forest Regressor was used, generating a new plot.



GRAPH 5

The new graph generated shows a more realistic forecast, presenting an annual growth projection with a short-term deceleration (2020-2027), tending to stabilize and begin a controlled recovery until 2030.

Unlike linear regression, we now have a realistically sustainable forecast for Brazilian e-commerce.

5.2. Final Recommendations

Having proven the effectiveness of the non-linear model in e-commerce data analysis and predictability, the suggestion for future and more robust analyses is the use of multiple non-linear models (such as XGBoost or Prophet), comparing the similarities between their projections.

Furthermore, it is crucial to establish a routine of continuous model validation with real data acquired annually, ensuring the model's adaptation to new macroeconomic and market dynamics.

6. REFERENCES

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7. APPENDIX

PROJECT GITHUB REPOSITORY: Predictive-Model-for-Magazine-Luiza-MGLU3-Revenue-Growth

This appendix details the structure of the project's public repository, where all source codes, data, and notebooks used for the analysis and predictive modeling presented in this report are stored.

The full code and dataset files can be accessed via the main repository link:

Access Link:

<https://github.com/elucas8013/Analise-Critica-da-Extrapolacao-Linear-no-E-commerce>

